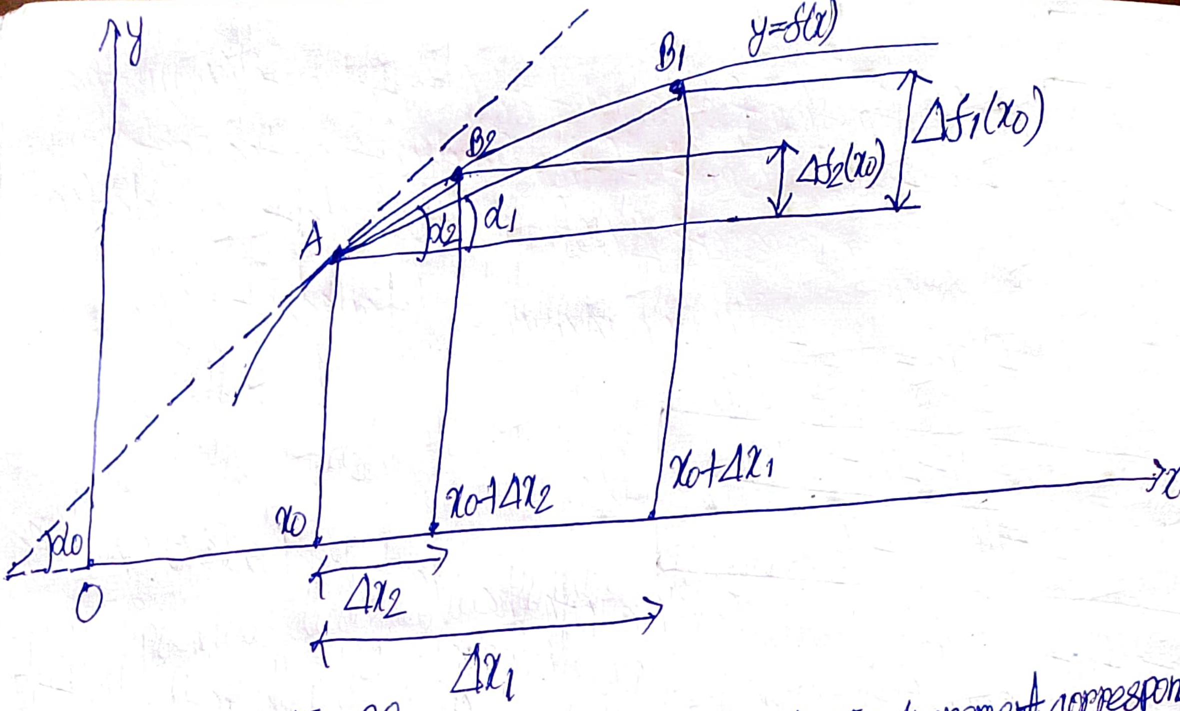

Fig. 38

Next take an increment Δx_2 (so that $\Delta x_2 < \Delta x_1$). This increment corresponds to the increment $\Delta f_2(x_0)$ of the function f at point x_0. Denote the slope of the chord AB_2 by α_2; it is similarly quite apparent that

$$\boxed{\frac{\Delta f_2(x_0)}{\Delta x_2} = \tan\alpha_2}$$

Further, take an increment Δx_3 ($\Delta x_3 < \Delta x_2$), and so on. As a result, we obtain an infinitesimal sequence of increments of the independent variable:

$$\Delta x_1, \Delta x_2, \Delta x_3, \ldots\ldots, \Delta x_n, \ldots\ldots$$

and the corresponding infinitesimal sequence of increments of the function at point x_0:

$$\Delta f_1(x_0), \Delta f_2(x_0), \Delta f_3(x_0), \ldots\ldots, \Delta f_n(x_0), \ldots\ldots$$

This leads to a new sequence of the values of the tangent of the slopes of the chords $AB_1, AB_2, AB_3, \ldots\ldots, AB_n, \ldots\ldots$ obtained as a sequence of the ratios of the two sequences given above

$$\boxed{\tan\alpha_1, \tan\alpha_2, \tan\alpha_3, \ldots\ldots, \tan\alpha_n, \ldots\ldots} \quad (3)$$

Both sequences (Δx_n) and $(\Delta f_n(x_0))$ converge to zero. And what can be said about the convergence of the sequence $(\tan\alpha_n)$, or, in other words, the sequence $\left(\frac{\Delta f_n(x_0)}{\Delta x_n}\right)$?